psgscoring

A Technical Handbook for Automated Polysomnography Scoring

Version 0.2.93 — April 2026

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`https://github.com/bartromb/psgscoring`

Abstract

This handbook provides a technical introduction to `psgscoring`, an open-source Python library (BSD-3) for AASM 2.6-compliant automated polysomnography (PSG) scoring. It is written for BSc-level computer science students with basic knowledge of Python, NumPy, and signal processing. The handbook covers the clinical context, signal processing pipeline, respiratory event detection algorithms, apnea type classification, correction mechanisms, **hypoxic burden computation**, and **post-processing for central apnea refinement**.

What's new in v0.2.93 (relative to v0.2.91):

- Hypoxic burden computation (Azarbarzin et al., AJRCCM 2019)
- Post-processing module: CSR-aware reclassification, mixed apnea decomposition, central instability index
- Pipeline expanded from 9 to 11 steps
- 42 public exports (was 38)
- External validation on iSLEEPS ($n = 96$, full dataset)

Prerequisites: Python 3.9+, NumPy, SciPy, basic DSP.
Repository: `https://github.com/bartromb/psgscoring`
Installation: `pip install psgscoring`

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1. Clinical Context: Why This Matters

1.1 Sleep Apnea

Obstructive Sleep Apnea (OSA) affects approximately 936 million adults worldwide. During sleep, the upper airway collapses partially or completely, causing breathing interruptions that fragment sleep and reduce blood oxygen levels. Left untreated, OSA increases the risk of hypertension, stroke, heart failure, and traffic accidents.

Diagnosis relies on polysomnography (PSG): an overnight recording of brain waves (EEG), eye movements (EOG), muscle tone (EMG), airflow, respiratory effort, blood oxygen saturation (SpO_2), body position, snoring, heart rhythm (ECG), and leg movements.

1.2 The Scoring Problem

A sleep technologist manually reviews the recording and counts:

- **Apneas**: ≥90% flow reduction for ≥10 seconds
- **Hypopneas**: ≥30% flow reduction for ≥10 seconds with ≥3% oxygen desaturation or an EEG arousal

The Apnea-Hypopnea Index (AHI) = (apneas + hypopneas) / hours of sleep. Manual scoring takes 1–2 hours per study with 10–20% inter-scorer variability. `psgscoring` automates this process.

1.3 Beyond AHI: The Pevernagie Critique

The AHI, though ubiquitous, has been critically appraised as a reductionist metric. Pevernagie et al. demonstrated that the correlation between AHI and key clinical endpoints—daytime sleepiness, hypertension, quality of life—is surprisingly weak [1]. Two patients with identical AHI values can experience fundamentally different physiological consequences.

`psgscoring` v0.2.93 addresses this by computing three complementary dimensions:

1. **AHI confidence interval** — event frequency with uncertainty
2. **Hypoxic burden** — oxygen deficit per hour (Section 4)
3. **Arousal index** — sleep fragmentation

2. Architecture Overview

`psgscoring` is organised as 13 independently importable submodules (12 in v0.2.91 + `postprocess` in v0.2.93):

Table 1: Submodule responsibilities and approximate size (v0.2.93)

Module	Responsibility	LOC
`constants`	AASM thresholds, band limits, scoring profiles	76
`utils`	Sleep mask construction, channel detection	134
`signal`	Linearisation, baseline, MMSD validation	327
`breath`	Breath segmentation, flattening index	254
`classify`	7-rule apnea type decision tree	310
`spo2`	SpO_2 coupling, ODI, **hypoxic burden**	395
`ecg_effort`	ECG-derived effort (TECG + spectral)	180
`plm`	Periodic Limb Movement detection	271
`ancillary`	Heart rate, snoring, position, CSR	277
`signal_quality`	Per-channel quality grading	160
`postprocess`	**CSR reclassification, mixed decomposition, CII**	**310**
`respiratory`	Apnea/hypopnea detection + 12 corrections	1061
`pipeline`	Master orchestrator (11 steps)	802

The public API exports 42 symbols from `psgscoring/__init__.py` (was 38 in v0.2.91).

3. Signal Processing Chain

The respiratory analysis transforms raw sensor data into events through six steps (unchanged from v0.2.91):

Step 1 — Nasal Pressure Linearisation. $x_{\text{lin}}(t) = \text{sign}(x(t)) \cdot \sqrt{|x(t)|}$ (Bernoulli correction).

Step 2 — Bandpass Filtering. 4th-order Butterworth, 0.05–3 Hz.

Step 3 — Hilbert Envelope. $A(t) = |x(t) + i\mathcal{H}\{x(t)\}|$.

Step 4 — Dynamic Baseline. 5-minute sliding 95th-percentile window, 10-second steps.

Step 5 — Event Detection. Threshold crossing: $A(t)/B(t) \leq 1 - \theta$ for $\geq$10 s.

Step 6 — MMSD Validation. Mean Magnitude of Second Derivative distinguishes true apneas from artefacts.

4. Hypoxic Burden (v0.2.93)

4.1 Clinical Motivation

The AHI counts events but ignores their physiological impact. A patient with 10 shallow desaturations (3%) and a patient with 10 deep desaturations (15%) have the same AHI but radically different cardiovascular risk. The hypoxic burden captures this by integrating the total oxygen deficit.

Azarbarzin et al. (2019) demonstrated that hypoxic burden independently predicts incident cardiovascular disease (HR 1.47 for the highest tertile) [3].

4.2 Algorithm

For each respiratory event i:

1. Compute the **pre-event SpO_2 baseline**: 90th percentile in a 120-second window before event onset, with a global 95th percentile fallback if the local window is insufficient.

2. **Integrate** the desaturation area:

$$\text{area}_i = \int_{t_{\text{onset}}}^{t_{\text{recovery}}} \max\big(0,\ B_{\text{SpO}_2} - \text{SpO}_2(t)\big)\, dt \tag{1}$$

where t_{recovery} is when $\text{SpO}_2 \geq B - 1\%$ or $t_{\text{onset}} + 120$ s (whichever comes first).

3. **Normalise**:

$$\text{Hypoxic burden} = \frac{\sum_i \text{area}_i}{60 \times \text{TST (hours)}} \quad [\% \cdot \text{min/h}] \tag{2}$$

4.3 Clinical Thresholds

Table 2: Hypoxic burden risk stratification (Azarbarzin et al. 2019)

Range	HB (%·min/h)	Interpretation
Low risk	<20	Low cardiovascular risk
Moderate	20–73	Moderate risk
High risk	>73	HR 1.47 for incident CVD

4.4 Code Example

```python
from psgscoring import compute_hypoxic_burden

result = compute_hypoxic_burden(
    spo2_data=spo2_array,     # raw SpO2 signal
    sf_spo2=1.0,              # sampling frequency (Hz)
    resp_events=events,        # from run_pneumo_analysis()
    hypno=hypno_list,          # epoch staging
)
print(f"HB: {result['hypoxic_burden']} {result['unit']}")
# Output: HB: 71.7 %*min/h
```

4.5 Validation Example: The Pevernagie Phenotype

On PSG-IPA, SN2 (AHI 8.9, Mild) has a hypoxic burden of 71.7 %·min/h—approaching the high-risk threshold. SN1 (AHI 7.6, also Mild) has HB 12.5. Same AHI category, 6× difference in oxygen deficit. On iSLEEPS ($n = 96$), 9 patients with AHI <15 had HB >40, including one with AHI 9.0 and HB 114.

5. Post-Processing Module (v0.2.93)

After the main detection pipeline, three post-processing refinements are applied automatically (Step 11):

5.1 CSR-Aware Central Reclassification

Problem: During Cheyne-Stokes respiration, cardiac pulsation on the effort bands can mimic respiratory effort, causing central apneas to be misclassified as obstructive.

Solution: Events with `csr_flagged=True` that are classified as `"obstructive"` or `"mixed"` are reclassified as `"central"`.

```python
from psgscoring import reclassify_csr_events

events = reclassify_csr_events(events, csr_info)
# Events now have: original_type, csr_reclassified=True
```

Validation: On PSG-IPA, 23 events were reclassified (7 in SN1, 16 in SN3). On iSLEEPS ($n = 96$), 365 events were reclassified across 36 studies—confirming that central apnea misclassification is the dominant error in stroke populations.

5.2 Mixed Apnea Decomposition

Problem: Mixed apneas with a long central portion (≥10 s) are physiologically more central than obstructive.

Solution: For each mixed apnea, the effort signal is analysed to measure the central (absent effort) and obstructive (resumed effort) portions. If the central portion ≥10 s, the event is reclassified as central.

```python
from psgscoring import decompose_mixed_apneas

events = decompose_mixed_apneas(
    events, thorax_data, abdomen_data, sf_effort,
    central_threshold_s=10.0,
)
# Each mixed event now has:
# central_duration_s, obstructive_duration_s, central_ratio
```

5.3 Central Instability Index (CII)

Problem: How uncertain is the obstructive/central classification?

Solution: Compare AHI across the three scoring profiles. If the profiles yield very different OAHI values, the O/C distinction is unstable.

```python
from psgscoring import compute_central_instability_index

cii = compute_central_instability_index(
    ahi_strict=4.3, ahi_standard=7.6, ahi_sensitive=5.5,
)
print(f"CII: {cii['central_instability_index']}")
print(f"     {cii['interpretation']}")
# CII: 0.43
#      high -- many ambiguous events, consider manual review
```

Table 3: Central Instability Index interpretation

CII	Range	Interpretation
Low	<0.15	Classification stable across profiles
Moderate	0.15–0.40	Some events are profile-sensitive
High	>0.40	Many ambiguous events; manual review

6. The Complete Pipeline (v0.2.93)

The master function `run_pneumo_analysis()` now orchestrates **11 steps** (was 9 in v0.2.91):

Table 4: Pipeline steps (v0.2.93)

Step	Description	New?
1	Signal quality assessment	
2	Linearisation + filtering + envelope	
3	Dynamic baseline computation	
4	Apnea detection (thermistor, ≥90%)	
5	Hypopnea detection (nasal pressure, ≥30%)	
6	SpO_2 coupling + type classification	
7	6 over-counting corrections	
8	6 under-counting corrections	
9	AHI confidence interval (3 profiles)	
10	**Hypoxic burden** (Azarbarzin 2019)	⋆
11	**Post-processing** (CSR recl. + mixed decomp.)	⋆

7. Updated Data Structures (v0.2.93)

7.1 Event Dictionary (additions)

```
{
    # ... existing fields from v0.2.91 ...
    # New in v0.2.93 (post-processing):
    "csr_reclassified": True,     # was obstructive, now central
    "original_type": "obstructive",
    "central_duration_s": 12.1,   # mixed decomposition
    "obstructive_duration_s": 7.9,
    "central_ratio": 0.60,
    "mixed_decomposed": True,
}
```

7.2 Hypoxic Burden Output

```
results["hypoxic_burden"] = {
    "hypoxic_burden": 71.7,          # %*min/h
    "total_area_pct_s": 25812.3,     # raw area in %*s
    "n_events_with_burden": 40,
    "mean_event_burden": 645.3,      # %*s per event
    "unit": "%*min/h",
}
```

7.3 Post-Processing Output

```
results["postprocess"] = {
    "n_csr_reclassified": 7,
    "n_mixed_decomposed": 3,
    "n_mixed_to_central": 2,
    "central_instability": {
        "central_instability_index": 0.43,
        "interpretation": "high",
        "ahi_range": 3.3,
    },
    "cai_change": +9,
}
```

8. External Validation (Updated)

8.1 PSG-IPA (v0.2.93)

Table 5: PSG-IPA results with v0.2.93 (5 recordings, 60 scorer sessions)

	n	$\mathbf{AHI_{med}}$	$\mathbf{AHI_{algo}}$	$\lvert\Delta\rvert$	**O**	**C**	**H**	**HB**	**CSR**
SN1	11	6.0	7.6	1.6	9	7	28	12.5	7
SN2	12	3.7	8.9	5.2	3	0	40	**71.7**	0
SN3	12	54.0	53.5	0.5	252	30	42	**241.8**	16
SN4	12	3.5	4.2	0.7	0	0	25	22.1	0
SN5	12	9.9	10.7	0.8	17	1	57	37.0	0
Mean				1.8					23 total

8.2 iSLEEPS ($n = 96$, v0.2.93)

Table 6: iSLEEPS validation by severity ($n = 96$ ischemic stroke patients)

	Overall ($n = 96$)	**Normal** ($n = 12$)	**Mild** ($n = 20$)	**Moderate** ($n = 26$)	**Severe** ($n = 38$)
Bias	−14.3/h	+1.3	−1.3	−12.4	−27.5
MAE	15.6/h	3.4	3.9	13.0	27.5
r	0.653				
Conc.	30%	83%	45%	23%	11%
CSR recl.	365				

Key findings:

- Normal/mild MAE 3.4–3.9/h (excellent, confirming v0.2.91 results)
- 365 CSR reclassifications in 36/96 studies (37%) — central apnea misclassification is the dominant error in stroke populations
- 9 patients with AHI <15 but HB >40 (the "Pevernagie phenotype")
- 15 patients with HB >73 (high CV risk)

9. Exercises (Updated)

8. **Hypoxic burden**: Generate a synthetic SpO_2 signal (baseline 95%) with three events: a shallow desaturation (3%), a moderate desaturation (8%), and a deep desaturation (15%). Call `compute_hypoxic_burden()` and verify that the deep event contributes the most to the total burden.

9. **CSR reclassification**: Create a list of 10 events, 5 marked as `csr_flagged=True` with type `"obstructive"`. Call `reclassify_csr_events()` and verify that all 5 are reclassified to `"central"` with `original_type="obstructive"`.

10. **Mixed decomposition**: Generate a 20-second effort signal with 12 s of silence (central) followed by 8 s of sinusoidal effort (obstructive). Call `decompose_mixed_apneas()` with `central_threshold_s=10.0` and verify the event is reclassified as central.

11. **Pevernagie phenotype**: Using the iSLEEPS validation CSV, identify all patients where AHI <15 but HB >40. How many of these patients would receive CPAP based on AHI alone? How does adding the hypoxic burden change the clinical picture?

References

[1] Pevernagie DA, et al. On the rise and fall of the apnea–hypopnea index: a historical review and critical appraisal. *J Sleep Res.* 2020;29(4):e13066.

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[4] Berry RB, et al. The AASM Manual for the Scoring of Sleep and Associated Events. Version 2.6. AASM; 2020.

[5] Vallat R, Walker MP. An open-source, high-performance tool for automated sleep staging. *eLife.* 2021;10:e70092.

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